A uniform bound for complete graph minors

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November 2026

Abstract

Graph Minors XIII states, as a result of Thomason, that for every integer $p \geq 1$ every non-null simple graph G with $|E(G)| \geq 2.68\, p\sqrt{\log_2 p}\,|V(G)|$ has a K_p minor. Thomason's paper gives the constant 2.68 only for large p. We prove that for every integer $p \geq 51$, every non-null simple graph G with $|E(G)| \geq (16/5)\, p\sqrt{\ln p}\,|V(G)|$ has a K_p minor. Since $(16/5)\sqrt{\ln 2} < 2.68$, this gives the statement of Graph Minors XIII for every $p \geq 51$. The proof uses Thomason's density reduction and a random partition of the vertex set into many small blocks at once. Two certificates, checked with exact rational arithmetic, cover $51 \leq p \leq 10^{44}$, and an analytic argument covers $\ln p \geq 100$.

1 Introduction

All graphs in this note are finite and simple. For a graph G we write $v(G) = |V(G)|$ and $e(G) = |E(G)|$. A graph is non-null if it has at least one vertex, and the density of a non-null graph G is $d(G) = e(G)/v(G)$. We write ln for the natural logarithm and $\log_2$ for the logarithm to base 2.

Graph Minors XIII [RS95, (5.5), page 83] states the following, as a result of Thomason [Tho84].

> **(5.5)** *Let $p \geq 1$ be an integer, and let G be a non-null simple graph with $|E(G)| \geq 2.68\, p(\log_2(p))^{1/2}|V(G)|$. Then G has a K_p-minor.*

Thomason's paper gives the constant 2.68 only asymptotically, that is, for all sufficiently large p [Tho84, page 264]. In this note we prove a bound of the same form that holds for every $p \geq 51$.

Main Theorem. *For every integer $p \geq 51$, every non-null graph G with $d(G) \geq \frac{16}{5}\, p\sqrt{\ln p}$ has a K_p minor.*

The Main Theorem gives (5.5) for every $p \geq 51$. Indeed, $\sqrt{\ln p} = \sqrt{\ln 2}\,\sqrt{\log_2 p}$, so it suffices that $(16/5)\sqrt{\ln 2} \leq 2.68 = 67/25$. We have $\ln 2 < 7/10$, because $e^{7/10} > 1 + 7/10 + (7/10)^2/2 + (7/10)^3/6 > 2$, and therefore

$$\left(\frac{16}{5}\right)^2 \ln 2 < \left(\frac{16}{5}\right)^2 \frac{7}{10} = 7.168 < 7.1824 = \left(\frac{67}{25}\right)^2.$$

Numerically, $(16/5)\sqrt{\ln 2} = 2.6641747557\ldots$, so that $(16/5)\, p\sqrt{\ln p} = 2.6641747557\ldots\, p\sqrt{\log_2 p}$.

For $1 \leq p \leq 50$, the statement (5.5) is proved in the Lean formalization only, by a different argument, and this note does not cover these cases. The note makes no claim about the constant 16/5 for $p \leq 50$.

Wood [Woo26, Theorem 1] proves an explicit bound of the same form for every $p \geq 2$: every graph of density at least $4.5\, p\sqrt{\ln p}$ has a K_p minor. His proof builds the branch sets one

after another, and it adds vertices to each branch set to make it connected. Here we choose all candidate branch sets at once, as the blocks of one random partition. We then discard the disconnected blocks, and enough further blocks to remove the missing adjacencies, and we bound the expected number of discarded blocks. No vertex is added to a block, so the proportion of nonneighbors of a vertex stays the same during the construction.

The proof goes as follows. Section 2 recalls Thomason's density reduction. It reduces the Main Theorem for p to finding a K_{p-2} minor in small graphs in which every vertex has few nonneighbors. Section 3 gives the tool for these graphs: a random partition of the vertices into blocks (Lemmas 1 and 2), with an explicit lower bound B on the expected number of blocks that form a complete minor. It remains to show that $B > p - 3$. Section 4 checks this for $51 \le p \le 109$ with a finite certificate, Section 5 for $110 \le p \le 10^{44}$ with a certificate that covers whole intervals of p at once, and Section 6 for $\ln p \ge 100$ by hand. Section 7 puts the three ranges together, and Section 8 describes how the Lean formalization uses this note. Appendix A proves the density reduction.

2 The density reduction

Let $\alpha_0 > 1$ be the solution of $\alpha_0 = 1 + \ln(2\alpha_0)$. It is unique, because the function $x - 1 - \ln(2x)$ has derivative $1 - 1/x > 0$ for $x > 1$, is negative at $x = 1$, and tends to infinity. We use the following form of Lemma 1 of Thomason [Tho84, page 262].

Thomason's lemma. *Let $p \ge 2$ and $c \ge 3$ be integers. Suppose that every non-null graph H with*

$$v(H) \le c \qquad \text{and} \qquad 2\delta(H) - v(H) \ge \lfloor c/\alpha_0 \rfloor - 3 \tag{1}$$

has a K_{p-2} minor. Then every non-null graph of density at least c has a K_p minor.

Thomason states the lemma with a real parameter r, and we take $r = c/\alpha_0$, so that $\alpha_0 r = c$ is an integer. The proof makes two neighborhood reductions, and the two vertices chosen in them give the two extra branch sets. Appendix A gives a complete proof of the lemma in the form above. In [Tho84] the lemma is stated in terms of an extremal function. Below we apply it only with an integer c strictly less than the density of the graph, so no question about a boundary case arises if one uses his formulation instead.

We do not need the value of α_0, only the bound $\alpha_0 < 67/25$. To prove it, recall that $x - 1 - \ln(2x)$ is strictly increasing for $x > 1$ and vanishes at α_0. At $x = 67/25$ it equals $42/25 - \ln(134/25)$, which is positive because

$$e^{42/25} > \sum_{j=0}^{7} \frac{(42/25)^j}{j!} = \frac{134}{25} + \frac{110787181}{30517578125} > \frac{134}{25}.$$

So $\alpha_0 < 67/25$. For an integer $c \ge 3$ put

$$R = \left\lfloor \frac{25c}{67} \right\rfloor - 1. \tag{2}$$

Let H satisfy (1), let $n = v(H)$, and let $q = n - 1 - \delta(H)$, which is the largest number of nonneighbors of a vertex of H. Here and below, a vertex is not counted as a nonneighbor of itself. Since $c/\alpha_0 > 25c/67$, the second condition in (1) implies $2\delta(H) - n \ge R - 2$. As $2\delta(H) - n = n - 2 - 2q$, this says that $n - 2q \ge R$. Together with $n \le c$ we get

$$R \le n \le c \qquad \text{and} \qquad 0 \le q \le \frac{n - R}{2}. \tag{3}$$

We call a non-null graph H with n vertices a *reduced graph* (for c) if $R \le n \le c$ and every vertex of H has at most $(n-R)/2$ nonneighbors. If $n = R$ or $n = R+1$, then $q = 0$, so a reduced graph of this order is complete.

Now let $p \ge 51$, and put

$$C = \frac{16}{5}, \qquad D = Cp\sqrt{\ln p}, \qquad c = \lceil D \rceil - 1. \tag{4}$$

Then $D - 1 \le c < D$, and $c \ge 3$ since $D > 4$. A graph as in the Main Theorem has density at least D, and so at least c. By Thomason's lemma and (3), the Main Theorem for p follows once every reduced graph for this c has a K_{p-2} minor. In Section 4 the certificate may instead give any integer $c \ge 3$ with $c < D$, and the same reduction applies. The bound $c \ge D - 1$ is not needed there.

3 A random partition into blocks

To find a large complete minor in a reduced graph H, we split a random set of vertices into blocks of k vertices each. Every vertex has few nonneighbors, so a random block is likely to be connected and to dominate almost all of H. Then two blocks are likely to be adjacent. We delete the blocks that fail, and the remaining blocks form a complete minor.

For a set S of vertices of a graph H, let $U(S) = V(H) \setminus N_H[S]$ be the set of vertices that are neither in S nor adjacent to a vertex of S. We say that S does not dominate these vertices. A K_m model in H is a family of m pairwise disjoint nonempty sets of vertices, each inducing a connected subgraph, with an edge between every two of them. A graph has a K_m minor if and only if it has a K_m model.

Lemma 1. *Let H be a graph with n vertices in which every vertex has at most q nonneighbors. Let k be an integer with $2 \le k < n$, put $N = \lfloor n/k \rfloor$ and $\pi = q/n$, and suppose that $N \ge 2$. Let $A > 0$, and put*

$$F = (2^{k-1} - 1)\left(\frac{q}{n-k+1}\right)^{k-1}, \qquad M = \binom{N}{2}\left(\frac{An\pi^k}{n-k}\right)^k,$$
$$B = N(1 - F - A^{-1}) - M. \tag{5}$$

If $B > m - 1$ for an integer m, then H has a K_m minor.

The three loss terms come from blocks that are disconnected (NF), blocks that do not dominate enough vertices (N/A), and pairs of blocks with no edge between them (M).

Proof. Take a uniformly random ordering of $V(H)$. Let S_1 be the set of its first k vertices, S_2 the set of the next k vertices, and so on up to S_N. We call these sets blocks. They are disjoint, and each one on its own is a uniformly random k-subset of $V(H)$.

Domination. Fix a block S_i and a vertex v. If $v \in U(S_i)$, then all k vertices of S_i are nonneighbors of v, and v has at most q nonneighbors. So

$$\mathbb{E}|U(S_i)| \le n\binom{q}{k}\Big/\binom{n}{k} \le n\pi^k,$$

where the last step uses $(q-j)/(n-j) \le q/n$ for $0 \le j < k$. Call S_i *good* if $|U(S_i)| \le An\pi^k$. If $q > 0$, Markov's inequality shows that S_i is not good with probability at most $1/A$. If $q = 0$, then $U(S_i)$ is empty, so every block is good. In both cases the expected number of blocks that are not good is at most N/A.

Connectivity. If $H[S_i]$ is disconnected, then there is a partition of the k positions of S_i into two nonempty parts with no edge of H between the vertices in the two parts. There are $2^{k-1}-1$ such partitions. Fix one, and fix a spanning tree T of the complete bipartite graph between its two parts. The tree T has $k-1$ edges, and if no edge of H goes across the partition, every edge of T joins two nonadjacent vertices of H. Root T, and reveal the vertices of S_i in an order of the positions in which every position comes after its parent. When a position other than the root is revealed, at least $n-k+1$ vertices are still available, all equally likely, and at most q of them are nonneighbors of the vertex at the parent position. So the probability that every edge of T joins nonadjacent vertices is at most $(q/(n-k+1))^{k-1}$. By the union bound over the partitions, $H[S_i]$ is disconnected with probability at most F, and the expected number of disconnected blocks is at most NF.

Missing pairs. Let $i<j$. Condition on S_i, and suppose that S_i is good. Given S_i, the block S_j is a uniformly random k-subset of the $n-k$ vertices outside S_i. If no edge joins S_i and S_j, then $S_j \subseteq U(S_i)$. Hence, as $|U(S_i)| \le n-k$,

$$\mathbb{P}(\text{no edge joins } S_i \text{ and } S_j \mid S_i) \le \binom{|U(S_i)|}{k} \Big/ \binom{n-k}{k} \le \left(\frac{|U(S_i)|}{n-k}\right)^k \le \left(\frac{An\pi^k}{n-k}\right)^k.$$

Let Z be the number of pairs $i<j$ such that S_i and S_j are both good and connected and no edge joins them. Dropping the conditions that S_j is good and that both blocks are connected can only increase the probability of such a pair. So the bound above, averaged over S_i and summed over the $\binom{N}{2}$ pairs, gives $\mathbb{E}Z \le M$.

The model. Delete every block that is not good or is disconnected. Then, for each pair of remaining blocks with no edge between them, delete one of the two blocks. This deletes at most Z further blocks. The remaining blocks are disjoint, each induces a connected subgraph, and every two of them are joined by an edge. So they form a K_t model in H, where t is their number. By linearity of expectation, $\mathbb{E}t \ge N - N/A - NF - M = B$. Hence some ordering gives $t \ge B > m-1$, and since t is an integer, $t \ge m$. So H has a K_m minor. □

The proof uses only the distribution of one block, or of one block given another, and linearity of expectation. It does not need any independence between the blocks.

3.1 The best choice of A

Lemma 1 holds for every $A>0$. For $q>0$ we choose A to make $N/A+M$ as small as possible. Write $M = c_0A^k$, with $c_0 = \binom{N}{2}(n\pi^k/(n-k))^k$. As a function of $A>0$, the sum $N/A + c_0A^k$ is smallest when $A^{k+1} = N/(kc_0)$, and then $c_0A^k = N/(kA)$, so that $N/A + M = (1+1/k)N/A$. Put $P = (1+1/k)/A$ for this A. Then Lemma 1 holds with

$$B = N(1-F-P), \qquad P^{k+1} = \left(1+\frac{1}{k}\right)^{k+1} \frac{k(N-1)}{2} \left(\frac{n}{n-k}\right)^k \pi^{k^2}. \tag{6}$$

This also holds for $q=0$, with $F=P=0$. Indeed, then $B = N(1-1/A)$ in Lemma 1, and if $N > m-1$, then $N(1-1/A) > m-1$ for every large enough A.

3.2 Blocks of sizes 2 and 3

For some cases of Section 4 it helps to use blocks of two sizes.

Lemma 2. *Let H be a graph with n vertices in which every vertex has at most q nonneighbors, and put $\pi = q/n$. Let $N_2, N_3 \geq 1$ be integers with $2N_2 + 3N_3 \leq n$, and let $A_2, A_3 > 0$. Put*

$$F_2 = \frac{q}{n-1}, \qquad F_3 = 3\left(\frac{q}{n-2}\right)^2,$$
$$M_{22} = \binom{N_2}{2}\left(\frac{A_2 n}{n-2}\right)^2 \pi^4, \qquad M_{33} = \binom{N_3}{2}\left(\frac{A_3 n}{n-3}\right)^3 \pi^9,$$
$$M_{23} = N_2 N_3 \left(\frac{A_2 n}{n-2}\right)^3 \pi^6,$$

and

$$B = N_2(1 - F_2 - A_2^{-1}) + N_3(1 - F_3 - A_3^{-1}) - M_{22} - M_{23} - M_{33}. \tag{7}$$

If $B > m - 1$ for an integer m, then H has a K_m minor.

Proof. We follow the proof of Lemma 1. In a uniformly random ordering of $V(H)$, the first $2N_2$ vertices form N_2 blocks of size 2, called pairs, and the next $3N_3$ vertices form N_3 blocks of size 3, called triples. A pair S is good if $|U(S)| \leq A_2 n\pi^2$, and a triple S is good if $|U(S)| \leq A_3 n\pi^3$. The domination and connectivity bounds of Lemma 1 with $k = 2$ and $k = 3$ show that the expected numbers of pairs and triples that are not good or are disconnected are at most $N_2(F_2 + 1/A_2)$ and $N_3(F_3 + 1/A_3)$. For two pairs, or for two triples, the bound on missing pairs of Lemma 1 gives the terms M_{22} and M_{33}. For a pair and a triple, condition on the pair and suppose that it is good. The triple is then a uniformly random 3-subset of the $n - 2$ other vertices, and it has no neighbor in the pair only if it lies in the set of vertices that the pair does not dominate. This happens with probability at most $(A_2 n\pi^2/(n-2))^3$, and summing over the $N_2 N_3$ choices of a pair and a triple gives M_{23}. The rest of the proof is the same as for Lemma 1. □

4 Exact certificate for $51 \leq p \leq 109$

In this range we check, for each p and each order n of a reduced graph, that Lemma 1 or Lemma 2 with $m = p - 2$ applies. Then every reduced graph has a K_{p-2} minor.

The file `global_finite_certificate.jsonl` has one row for each integer p from 51 to 109. A row gives an integer c and one choice of blocks for each order

$$n = R + 2, R + 3, \ldots, c, \qquad \text{where } R = \lfloor 25c/67 \rfloor - 1.$$

The checker `global_finite_checker.py` first verifies, with a rational lower bound ℓ_p on $\ln p$ (Section 5.1), that

$$c^2 < C^2 p^2 \ell_p \qquad \text{and} \qquad R \geq p - 2.$$

The first inequality gives $c < D$. The second gives $c \geq 3$, since it needs $\lfloor 25c/67 \rfloor \geq 50$, and it handles every complete reduced graph, since such a graph has at least $R \geq p - 2$ vertices. By (3), this includes every reduced graph of order $n = R$ or $n = R + 1$. For each $n \geq R + 2$ the checker uses the bound

$$q = \left\lfloor \frac{n - R}{2} \right\rfloor,$$

the largest number of nonneighbors of a vertex that a reduced graph of order n allows. Lemmas 1 and 2 need only an upper bound q on the number of nonneighbors of each vertex, so a check for this value of q covers every reduced graph of order n. (Equivalently, every loss term in (5) and (7) is nondecreasing in q.)

A uniform entry of a row gives an integer k and a positive rational A, for Lemma 1. A mixed entry gives integers N_2, N_3 and positive rationals A_2, A_3, for Lemma 2. For each entry the

checker verifies the conditions of the lemma, computes B from (5) or (7) with exact fractions, and verifies $B > p - 3$. It also verifies that the rows cover every p from 51 to 109, and that each row covers every n from $R + 2$ to c. The result is as follows.

Quantity	Verified value
Values of p	$51, 52, \ldots, 109$
Cases (p, n) checked	19 873
Cases using Lemma 2	313
Lower bound for every $B - (p - 3)$	1/10 (strict)

The smallest margin $B - (p - 3)$ is about 0.100642159415. The checker verifies with exact fractions that every margin is greater than 1/10. So Lemma 1 or Lemma 2 gives a K_{p-2} minor in every reduced graph for $51 \le p \le 109$.

5 Exact interval certificate for $110 \le p \le 10^{44}$

For larger p we cannot check each value of p on its own. So the second certificate covers whole intervals of p at once. Within an interval, it also covers the order n of a reduced graph by a few ranges, after dividing n by D. Every check is again exact.

Fix integers $110 \le L \le U$, and let s_L and s_U be rational numbers with

$$s_L \le \sqrt{\ln L} \qquad \text{and} \qquad s_U \ge \sqrt{\ln U}.$$

Section 5.1 explains how they are computed. Let $D_L = CLs_L$, $c_L = \lceil D_L \rceil - 1$ and $R_L = \lfloor 25c_L/67 \rfloor - 1$. Now let p be an integer with $L \le p \le U$, and let D, c and R be as in (4) and (2). Then $CLs_L \le D \le CUs_U$, because $s_L \le \sqrt{\ln L} \le \sqrt{\ln p} \le \sqrt{\ln U} \le s_U$, and $c \ge 3$ since $D > 4$.

We first bound R/D from below. Write $\bar{\alpha} = 67/25$. Since $\lfloor y \rfloor > y - 1$ for every real y, and $c \ge D - 1$,

$$R > \frac{c}{\bar{\alpha}} - 2 \ge \frac{D}{\bar{\alpha}} - \left(\frac{1}{\bar{\alpha}} + 2\right).$$

Dividing by $D \ge D_L$ gives the first bound below. Also $c \ge c_L$, because $D \ge D_L$, and hence $R \ge R_L$. With $D \le CUs_U$, this gives the second bound below when $R_L \ge 0$. So

$$\frac{R}{D} \ge \rho := \max\left\{\frac{1}{\bar{\alpha}} - \frac{1/\bar{\alpha} + 2}{D_L},\ \frac{R_L}{CUs_U}\right\}.$$

The checker verifies $\rho > 0$. If the second term is the larger one, it is positive, so $R_L > 0$ and the second bound applies.

Let H be a reduced graph with n vertices, let q be the largest number of nonneighbors of a vertex of H, and put $\pi = q/n$ and $x = n/D$. By (3), $R \le n \le c < D$, so $\rho \le x \le 1$.

The certificate splits $[\rho, 1]$ into intervals $[a, b]$ with rational ends. We call an interval of p together with one interval of x a *cell*. For a cell put

$$n_- = CLs_L a, \qquad n_+ = CUs_U b, \qquad \pi_+ = \frac{1 - \rho/b}{2}.$$

If $x \in [a, b]$, then $n_- \le n \le n_+$, since $n = xD$. Also $\pi \le \pi_+$, because $q \le (n - R)/2$ by (3) and $R/n = (R/D)/x \ge \rho/b$. Each cell gives an integer $k \ge 2$ and a rational $P_+ \ge 0$. The checker verifies $n_- > k$, $n_-/k \ge 2$ and $0 \le \pi_+ < 1/2$, so that $2 \le k < n$ and $N = \lfloor n/k \rfloor \ge 2$, as Lemma 1 requires.

We use Lemma 1 in the form (6), which also covers $q = 0$. Since $q/(n-k+1) = \pi/(1-(k-1)/n)$, the connectivity loss satisfies $F \le F_+$, where

$$F_+ = (2^{k-1} - 1)\left(\frac{\pi_+}{1-(k-1)/n_-}\right)^{k-1}.$$

In (6) we have $k(N-1) \le n \le n_+$, and $n/(n-k)$ decreases as n increases, so $n/(n-k) \le n_-/(n_- - k)$. Hence $P^{k+1} \le Y$, where

$$Y = \left(1+\frac{1}{k}\right)^{k+1} \frac{n_+}{2}\left(\frac{n_-}{n_- - k}\right)^k \pi_+^{k^2}.$$

The checker verifies $P_+^{k+1} \ge Y$, which gives $P \le P_+$ without computing any root. It also verifies $F_+ + P_+ < 1$ and

$$\left(\frac{Cs_La}{k} - \frac{1}{L}\right)(1 - F_+ - P_+) > 1 - \frac{3}{U}. \tag{8}$$

This gives $B > p - 3$, as follows. Since $N > n/k - 1$, $n \ge aD$ and $D \ge Cps_L$, we have

$$\frac{N}{p} > \frac{n}{kp} - \frac{1}{p} \ge \frac{Cs_La}{k} - \frac{1}{L}.$$

As $1 - F - P \ge 1 - F_+ - P_+ > 0$, this and (8) give

$$\frac{B}{p} = \frac{N}{p}(1-F-P) > \left(\frac{Cs_La}{k} - \frac{1}{L}\right)(1 - F_+ - P_+) > 1 - \frac{3}{U} \ge 1 - \frac{3}{p},$$

which is $B > p - 3$.

The file `global_medium_certificate.jsonl` has 88 intervals of p and 508 cells. The checker `global_medium_checker.py` verifies that the first interval starts at 110, that each interval starts one after the end of the previous one, and that the last one ends at 10^{44}. Within each interval it verifies that the intervals of x start at the computed ρ, that each starts where the previous one ends, and that the last one ends at 1. Every cell passes (8). The smallest margin of the left side of (8) above $1 - 3/U$ is about $9.98234667206 \times 10^{-7}$, and the checker verifies with exact fractions that every margin is positive.

5.1 Exact bounds on logarithms and square roots

For a rational number z with $0 \le z < 1$ and for $J = 24$, put

$$T_J(z) = 2\sum_{j=0}^{J-1} \frac{z^{2j+1}}{2j+1}.$$

The series $2\,\mathrm{atanh}\, z = \sum_{j\ge 0} 2z^{2j+1}/(2j+1)$ has positive terms, and its tail after J terms is at most the geometric series $\sum_{j\ge J} 2z^{2j+1}/(2J+1)$. So

$$T_J(z) \le 2\,\mathrm{atanh}\, z \le T_J(z) + \frac{2z^{2J+1}}{(2J+1)(1-z^2)}. \tag{9}$$

For an integer $m \ge 1$ let $h = \lfloor \log_2 m \rfloor$, which is one less than the number of binary digits of m, and put $z = (m - 2^h)/(m + 2^h)$. Then $0 \le z < 1/3$, and

$$\ln m = h \ln 2 + 2\,\mathrm{atanh}\, z, \qquad \ln 2 = 2\,\mathrm{atanh}(1/3),$$

because $2\operatorname{atanh} z = \ln((1+z)/(1-z))$. Applying (9) to both terms gives rational bounds $\ell_m \le \ln m \le u_m$. The finite checker uses only the lower bound.

For the interval checker, let $S = 10^{12}$, and compute with integer square roots

$$s_m^- = \frac{\left\lfloor \sqrt{\lfloor S^2 \ell_m \rfloor} \right\rfloor}{S}, \qquad s_m^+ = \frac{\left\lfloor \sqrt{\lfloor S^2 u_m \rfloor} \right\rfloor + 1}{S}.$$

Then $s_m^- \le \sqrt{\ell_m} \le \sqrt{\ln m}$, and $s_m^+ > \sqrt{u_m} \ge \sqrt{\ln m}$ because $(S s_m^+)^2 \ge \lfloor S^2 u_m \rfloor + 1 > S^2 u_m$. The checker takes $s_L = s_L^-$ and $s_U = s_U^+$. All quantities that the checkers accept are computed with integer or rational arithmetic. The scripts also contain generators, which use floating point to propose certificates. The checkers do not rely on the generators.

6 An analytic proof for $\ln p \ge 100$

For very large p we show $B > p - 3$ by hand, with $A = 4$. The block size k grows like $\sqrt{\ln p}$, which makes the connectivity loss and the missing-pair loss small. Put $s = \sqrt{\ln p} \ge 10$, so that $p = e^{s^2}$. Let H be a reduced graph with n vertices, and let q be the largest number of nonneighbors of a vertex of H. With c and R from (4) and (2),

$$R \ge \frac{80}{67} ps - \frac{159}{67} \ge 1.19ps, \qquad n \le 3.2ps. \tag{10}$$

The first inequality holds because $R > 25c/67 - 2$ and $c \ge D - 1$. The second follows from $80/67 - 119/100 = 27/6700$ and $ps \ge 600$, which holds as $p \ge e^{100}$. The bound on n holds because $n \le c < D$.

If $q = 0$, then H is complete and $n \ge R > p - 2$, so H has a K_{p-2} minor. Suppose $q > 0$, and let

$$\pi = q/n \qquad \text{and} \qquad \lambda = \ln(1/\pi).$$

By (3) and (10),

$$0 < \pi \le \frac{1 - R/n}{2} \le \frac{1 - 1.19/3.2}{2} = \frac{201}{640} < \frac{1}{3}, \qquad \text{so} \qquad \lambda > \ln 3 > 1.09.$$

Here $\ln 3 = 2\operatorname{atanh}(1/2) > 1 + 1/12 + 1/80 + 1/448 = 7379/6720 > 1.09$, from the first four terms of the series. Choose

$$k = \left\lceil \frac{s}{\sqrt{\lambda}} + 1 \right\rceil, \qquad N = \lfloor n/k \rfloor, \qquad A = 4.$$

Since $\lambda > 1$, we have $2 \le k < s/\sqrt{\lambda} + 2 < s + 2$. So, by (10) and $s/(s+2) \ge 10/12$,

$$\frac{n}{k} > \frac{1.19ps}{s+2} \ge \frac{119}{120} p > 1000.$$

In particular $N \ge 2$, $k/n < 1/1000$ and $n/(n-k) < 1000/999$. As $e^{7/10} > \sum_{j=0}^{3} (7/10)^j/j! > 2000/999$ and $e^{7/5} > \sum_{j=0}^{5} (7/5)^j/j! > 4000/999$, we get

$$\ln \frac{2n}{n-k+1} < 0.7, \qquad \ln \frac{4n}{n-k} < 1.4. \tag{11}$$

6.1 Connectivity loss

The loss F of Lemma 1 satisfies

$$\begin{aligned} F &< \left(\frac{2q}{n-k+1}\right)^{k-1} < \exp(-(k-1)(\lambda-0.7)) \\ &\le \exp\left[-s\left(\sqrt{\lambda}-\frac{0.7}{\sqrt{\lambda}}\right)\right] < e^{-0.37s} \le e^{-3.7} < 0.03. \end{aligned}$$

The first step uses $2^{k-1}-1<2^{k-1}$. The second uses $2q/(n-k+1)=\pi\cdot 2n/(n-k+1)$ and (11). The third uses $k-1\ge s/\sqrt{\lambda}$ and $\lambda>0.7$. For the fourth, the function $\sqrt{\lambda}-0.7/\sqrt{\lambda}$ increases with λ, and at $\lambda=1.09$ it equals $0.39/\sqrt{1.09}>0.37$, since $0.39^2=0.1521>0.149221=0.37^2\cdot 1.09$. The last two steps use $s\ge 10$ and $e^{3.7}>\sum_{j=0}^{5}3.7^j/j!>100/3$.

6.2 Missing-pair loss

With $A=4$, the term M of Lemma 1 and (11) give

$$\frac{M}{N}=\frac{N-1}{2}\left(\frac{4n\pi^k}{n-k}\right)^k \le \frac{N-1}{2}\exp(1.4k-k^2\lambda) \le 1.6ps\exp(1.4k-k^2\lambda),$$

where the last step uses $N-1<n\le 3.2ps$. The function $y^2\lambda-1.4y$ is increasing for $y\ge 0.7/\lambda$, and $k\ge s/\sqrt{\lambda}+1>0.7/\lambda$. Hence

$$\begin{aligned} k^2\lambda-1.4k &\ge \left(\frac{s}{\sqrt{\lambda}}+1\right)^2\lambda-1.4\left(\frac{s}{\sqrt{\lambda}}+1\right)=s^2+\left(2\sqrt{\lambda}-\frac{1.4}{\sqrt{\lambda}}\right)s+\lambda-1.4 \\ &> s^2+0.74s-0.31, \end{aligned}$$

where the last step uses $2\sqrt{\lambda}-1.4/\sqrt{\lambda}>2\cdot 0.37$ from the previous subsection and $\lambda>1.09$. Since $p=e^{s^2}$,

$$\frac{M}{N}<1.6s\exp(-0.74s+0.31)<0.02.$$

The middle expression decreases for $s\ge 10$, since the derivative of $se^{-0.74s}$ is $(1-0.74s)e^{-0.74s}$. At $s=10$ it equals $16e^{-7.09}$, which is less than 0.02 because $e^{7.09}>\sum_{j=0}^{8}7.09^j/j!>800$. Combining the two losses with $A^{-1}=1/4$ gives

$$B>N(1-0.03-0.25-0.02)=0.70N. \tag{12}$$

6.3 There are enough blocks

For $0<\pi\le 1/2$,

$$\frac{\mathrm{d}}{\mathrm{d}\pi}\left[\ln(1/\pi)-2(1-2\pi)^2\right]=-\frac{1}{\pi}+8(1-2\pi)=-\frac{(4\pi-1)^2}{\pi}\le 0.$$

So the bracket is at least its value $\ln 2$ at $\pi=1/2$, and

$$\lambda\ge 2(1-2\pi)^2+\ln 2\ge(2+\ln 2)(1-2\pi)^2,$$

where the last step uses $(1-2\pi)^2\le 1$. The first two terms of the series for $2\,\mathrm{atanh}(1/3)$ give $\ln 2>56/81>2/3$, and $0.613^2=0.375769>3/8$. Hence

$$\frac{1-2\pi}{\sqrt{\lambda}}\le\frac{1}{\sqrt{2+\ln 2}}<\sqrt{3/8}<0.613.$$

Now $n(1-2\pi) = n - 2q \geq R \geq 1.19ps$ by (3) and (10), and $k < s/\sqrt{\lambda} + 2$. So

$$\frac{n}{k} > \frac{1.19ps}{(1-2\pi)(s/\sqrt{\lambda}+2)} = \frac{1.19p}{(1-2\pi)/\sqrt{\lambda} + 2(1-2\pi)/s} \geq \frac{1.19p}{0.613 + 2/s} \geq \frac{1.19p}{0.813} > 1.46p.$$

Thus $N > n/k - 1 > 1.46p - 1$, and (12) gives

$$B > 0.70(1.46p - 1) = 1.022p - 0.7 > p - 3.$$

So Lemma 1 with $m = p - 2$ gives a K_{p-2} minor in H.

7 Proof of the Main Theorem

Let $p \geq 51$. By Section 2, it suffices to show that every reduced graph has a K_{p-2} minor, for the c of (4), or for the c of the certificate when $p \leq 109$. The three ranges are covered as follows.

Range	Argument
$51 \leq p \leq 109$	19 873 exact checks (Section 4)
$110 \leq p \leq 10^{44}$	88 intervals of p and 508 exact cells (Section 5)
$\ln p \geq 100$	analytic argument (Section 6)

The interval checker also verifies $\ell_m > 100$ for $m = 10^{44}$, with the lower bound of Section 5.1. So $\ln p > 100$ for every $p > 10^{44}$, and the three ranges cover every $p \geq 51$. In each case Lemma 1 or Lemma 2 gives a K_{p-2} model in every reduced graph, and Thomason's lemma gives the Main Theorem. Together with the Lean proof for $p \leq 50$ (Section 8), this proves (5.5) of Graph Minors XIII for every $p \geq 1$, with no unspecified threshold on p.

The two checkers and their certificates are in the folder `docs/thomason` of the GraphMinor repository (`https://github.com/mahdi-jfri/GraphMinor`), next to this note. They need Python 3.8 or later and no other package. The commands

```
python3 global_finite_checker.py check global_finite_certificate.jsonl
python3 global_medium_checker.py check global_medium_certificate.jsonl
```

run the checks of Sections 4 and 5, and `python3 verify_all.py` runs both. The certificate files contain only witnesses, that is, the choices of blocks and parameters. The checkers verify the coverage of the ranges and every inequality that the proof uses.

8 The Lean formalization

The Lean formalization proves (5.5) of Graph Minors XIII for every $p \geq 1$, as the theorem `GraphMinor.Thomason.thomason84` in `GraphMinor/Thomason/UniformTheorem.lean`. It splits the proof into four ranges of p. All file names below are in the folder `GraphMinor/Thomason`.

The range $p \leq 50$. The proof is in `FiftyCase.lean` and the files it imports. It does not follow this note.

The density reduction. Thomason's lemma is proved in `DensityReduction.lean`, following [Tho84], for every real $a > 1$ with $1+\ln(2a) \leq a$ in place of α_0. The file `UniformReduction.lean` applies it with $a = 67/25$, which satisfies this condition by the bound $e^{42/25} > 134/25$ of Section 2. It gives the reduction to reduced graphs of Section 2 with $R = \lfloor 25c/67 \rfloor - 1$.

The random partition. Lemma 1 is proved in `UniformPartition.lean`, in a form that allows blocks of different sizes. The expectations of the proof are written as averages over all injective maps from the positions of the blocks to the vertices (`UniformSamples.lean`, `UniformBlockCounts.lean`, `UniformBlockSums.lean` and `UniformPairSums.lean`). The file `UniformTwoSizes.lean` specializes it to blocks of two sizes, which covers Lemma 1 and Lemma 2, and `UniformCheck.lean` turns its hypothesis into a test on natural numbers that Lean's kernel can evaluate.

The range $51 \le p \le 109$. The Lean proof uses the certificate of Section 4. The script `gen_lean_finite.py` copies the value of c and the block choices of each row of `global_finite_certificate.jsonl` unchanged into the files `UniformFinite51.lean` to `UniformFinite109.lean`, and the kernel checks each row (`rowOK` in `UniformRows.lean`). The only difference is the density test: the kernel checks $c \le 2.68\,p\sqrt{\log_2 p}$ directly, with integers a and $b = 1000$ such that $2^a \le p^b$ and $10000\,bc^2 \le 71824\,ap^2$, instead of $c < D$. The file `UniformFiniteRange.lean` collects the rows.

The range $110 \le p \le 10^{44}$. Here the Lean proof uses its own certificate in base 2, not `global_medium_certificate.jsonl`. It keeps the density $D = 2.68\,p\sqrt{\log_2 p}$ of (5.5) itself, with $c = \lfloor D \rfloor$, and follows the method of Section 5. The logarithms are bounded by the integer comparisons $2^{a_L} \le L^{1000}$ and $U^{1000} \le 2^{a_U}$, and each cell also gives a rational upper bound on π. Instead of the best A of (6), each cell uses $A = (k+1)/(kP_+)$, which gives the same bound on the losses and needs no root. The certificate is `medium_log2_certificate.jsonl`, with 69 intervals and 261 cells, made and checked by `medium_log2_certificate.py`. The script `gen_lean_medium.py` writes it into the files `UniformMedium0.lean` to `UniformMedium6.lean` and `UniformMediumRange.lean`. The kernel checks each interval (`intervalOK` in `UniformIntervalCheck.lean`), and `UniformCell.lean` and `UniformIntervals.lean` prove that an accepted interval gives a K_{p-2} minor in every reduced graph.

The range $p > 10^{44}$. The files `UniformTail.lean` and `UniformLarge.lean` replace Section 6 by an argument in base 2. With $s = \sqrt{\log_2 p} \ge 12$, so that $p = 2^{s^2}$, they split the range of $\pi = q/n$ into the three pieces $\pi \le 1/25$, $1/25 < \pi \le 13/100$ and $13/100 < \pi \le 63/200$. On each piece they take blocks of size $\lceil ts \rceil + 1$, for a fixed t that depends on the piece, and $A = 4$. The connectivity loss and the missing-pair loss per block are then each at most $1/50$, and there are more than $1.41p - 1$ blocks, so Lemma 1 applies with $B > 0.71(1.41p - 1) > p - 3$.

A Proof of Thomason's lemma

Let $p \ge 2$ and $c \ge 3$ be integers, put $r = c/\alpha_0$, and suppose that every non-null graph satisfying (1) has a K_{p-2} minor. Since $\alpha_0 < 67/25$, we have $r > 1$. Let G be a non-null graph with $e(G) \ge c\,v(G)$. We find a K_p minor in two steps. Each step passes to a minimal minor and then to the neighborhood of a vertex of minimum degree. That vertex then gives one more branch set.

Step 1. Among the non-null minors G_1 of G with $e(G_1) \ge c\,v(G_1)$, choose one with $v(G_1)+e(G_1)$ as small as possible. Then no deletion of a vertex, deletion of an edge, or contraction of an edge of G_1 gives another such minor. We have $e(G_1) = c\,v(G_1)$, since otherwise $e(G_1) \ge c\,v(G_1) + 1$, as c is an integer, and deleting an edge would keep the inequality. As $c\,v(G_1) = e(G_1) \le v(G_1)(v(G_1)-1)/2$, we have $v(G_1) \ge 2c+1$. Let z be a vertex of minimum degree in G_1, and let $H_2 = G_1[N(z)]$. Then $v(H_2) = \deg z \le 2e(G_1)/v(G_1) = 2c$. Deleting z leaves a graph with $v(G_1)-1$ vertices and $e(G_1) - \deg z$ edges, so minimality gives $\deg z \ge c+1$. For each neighbor w of z, contracting zw removes one vertex and exactly $1 + |N(z) \cap N(w)|$ edges, so minimality gives $e(G_1) - 1 - |N(z) \cap N(w)| < c(v(G_1)-1)$, that is, $|N(z) \cap N(w)| \ge c$. So every vertex of

H_2 has degree at least c in H_2, and

$$c+1 \le v(H_2) \le 2c, \qquad \delta(H_2) \ge c.$$

Since z is adjacent to every vertex of H_2, a K_{p-1} model in H_2 together with $\{z\}$ is a K_p model in G_1.

Step 2. For real $x \ge r$ put

$$\Psi(x) = \frac{rx}{2}\left(\ln\frac{x}{r}+1\right).$$

This function is increasing, since its derivative is $(r/2)(\ln(x/r)+2) > 0$. For $r \le x \le 2c$ we have $\Psi(x) \le cx/2$, because $\ln(x/r) \le \ln(2\alpha_0) = \alpha_0 - 1$ by the definition of α_0, so that $r(\ln(x/r)+1) \le \alpha_0 r = c$. As $r < c+1 \le v(H_2) \le 2c$, this gives $\Psi(v(H_2)) \le c\,v(H_2)/2 \le e(H_2)$, where the last step uses $\delta(H_2) \ge c$.

Among the minors J of H_2 with $v(J) \ge r$ and $e(J) \ge \Psi(v(J))$, choose one with $v(J)+e(J)$ as small as possible, and put $n = v(J)$. Such minors exist, since H_2 itself is one. No graph with $r \le n < r+1$ has these properties, because then

$$\binom{n}{2} < \frac{rn}{2} \le \Psi(n).$$

So $n \ge r+1$, and deleting a vertex or contracting an edge of J leaves a graph with at least r vertices. Since $e(J)$ is an integer and deleting an edge does not change n, minimality gives $e(J) = \lceil \Psi(n) \rceil$. The graph J has no isolated vertex, because deleting one would keep $e(J) \ge \Psi(n) \ge \Psi(n-1)$, as Ψ is increasing.

Let u be a vertex of minimum degree in J, and let $H = J[N_J(u)]$. Then H is non-null, since u is not isolated. Since $n \ge r+1 > 2$ and $n \le v(H_2) \le 2c$,

$$v(H) = \deg u \le \frac{2\lceil \Psi(n) \rceil}{n} < \frac{2\Psi(n)+2}{n} \le r\left(\ln\frac{n}{r}+1\right)+1.$$

As $v(H)$ is an integer, this gives

$$v(H) \le \left\lceil r\left(\ln\frac{n}{r}+1\right)\right\rceil \le \lceil \alpha_0 r \rceil = c,$$

where the second step uses $n \le 2c$, as in the bound $\Psi(x) \le cx/2$ above. For each neighbor w of u, contracting uw removes one vertex and $1 + |N_J(u) \cap N_J(w)|$ edges, so minimality gives

$$e(J) - 1 - |N_J(u) \cap N_J(w)| < \Psi(n-1).$$

The degree of w in H is $|N_J(u) \cap N_J(w)|$, so $\delta(H) > \Psi(n) - \Psi(n-1) - 1$. With the bound on $v(H)$ above,

$$\begin{aligned} 2\delta(H) - v(H) &> 2\Psi(n) - 2\Psi(n-1) - r\left(\ln\frac{n}{r}+1\right) - 3 \\ &= r(n-1)\ln\frac{n}{n-1} - 3 > r - 4. \end{aligned}$$

The equality follows from the definition of Ψ. The last step uses $\ln(n/(n-1)) > 1/n$, which gives $r(n-1)\ln(n/(n-1)) > r - r/n > r-1$, as $n > r$. Since $2\delta(H) - v(H)$ is an integer, it is at least $\lfloor r \rfloor - 3 = \lfloor c/\alpha_0 \rfloor - 3$. So H satisfies (1), and by assumption H has a K_{p-2} model. Adding $\{u\}$ to it gives a K_{p-1} model in J. Since J is a minor of H_2, the graph H_2 has a K_{p-1} minor, and by Step 1 the graph G_1 has a K_p minor. As G_1 is a minor of G, so does G.

References

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